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OBJECTIVE

- To enhance and broaden my knowledge and contribute with the same to the scientific community and society.
- To help nourish upcoming scientific brains and direct them towards better utilization of their own capabilities.

EDUCATION

- 2021 – ... ■ **Doctor of Philosophy (Ph.D.) — Computer Science and Engineering.**
- 1 Indian Institute of Technology Guwahati, Assam, India, jointly with
 - 2 Indian Institute of Technology (BHU) Varanasi, Uttar Pradesh, India.
- Coursework grade: 8.75/10.00
Research area: *Designing Machine Learning-based Security Scheme for Inside and Outside Mimicry Attack Detection in Industrial Internet of Things (IIoT) Networks.*
- 2019 – 2021 ■ **Master of Technology (M.Tech.) — Computer Science and Engineering.**
National Institute of Technology Durgapur (NIT Durgapur), West Bengal, India.
Final grade: 9.28/10.00
Thesis title: *A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond.*
- 2006 – 2010 ■ **Bachelor of Engineering (B.E.) — Computer Science and Engineering.**
**University Institute of Technology (UIT),
The University of Burdwan (BU), West Bengal, India.**
Final grade: 79.9%
- 2004 – 2006 ■ **CBSE AISSE (Class XII) — Physics, Chemistry, Mathematics, Biology, English.**
Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
Final grade: 79.6%
- ... – 2004 ■ **CBSE AISSE (Class X).**
Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
Final grade: 82.8%

EMPLOYMENT HISTORY

- 2011 – 2019 ■ **Bio-Statistical Programmer, Data Modeler, and Data Analyst.**
Tata Consultancy Services (TCS), Delhi-NCR, India.
- 1 I have worked for a Danish Pharmaceutical Client* in the Lifesciences domain as a SAS Biostatistical programmer doing data analysis and interpretation to establish the safety and efficacy of upcoming drugs undergoing clinical trials in phases 1, 2, and 3.
 - 2 I have also worked for an American Insurance Client* in the Healthcare industry for health-related data modeling and trend prediction of US citizens.

* Names of the organizations have not been disclosed due to contractual obligations.

RESEARCH PUBLICATIONS

JOURNAL ARTICLES

- 1 **Sanyal, S., Khatua, M., & Chattopadhyay, P. (2026b).** LITMUS: A Lightweight Collaborative Intrusion Detection System for Host Oriented Mimicry Attack Detection in Industrial Internet of Things Networks with Dishonest Collaborators in the Majority. *Computer Networks*, 286, 1–28. (In Press). [doi:10.1007/978-3-032-22798-xxxx](https://doi.org/10.1007/978-3-032-22798-xxxx)
- 2 **Sanyal, S., Khatua, M., & Chattopadhyay, P. (2026c).** CIPHER: A Collaborative IDS Placement and Scheduling Framework for Host-Oriented Mimicry Attack Detection in IoT Networks. *IEEE Internet of Things Journal*. [doi:10.1109/JIOT.2026.3689273](https://doi.org/10.1109/JIOT.2026.3689273)
- 3 **Sanyal, S., Mukhopadhyay, S., & Xhafa, F. (2023).** Enhanced incentive schemes for volunteers to distribute surplus food in smart cities. *Sustainable Cities and Society*, 91, 104409. [doi:https://doi.org/10.1016/j.scs.2023.104409](https://doi.org/10.1016/j.scs.2023.104409)
- 4 **Sanyal, S., Kumar Singh, V., Xhafa, F., Sanyal, B., & Mukhopadhyay, S. (2021).** A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond. *Applied Sciences*, 11(11), 5058. [doi:10.3390/app11115058](https://doi.org/10.3390/app11115058)

CONFERENCE PROCEEDINGS

- 1 **Sanyal, S., Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026a, July).** A Survey of Project Selection in Tourist Spots using Smart City Platforms. In *the 20th International Conference on Complex, Intelligent, and Software Intensive Systems (CISIS-2026)* (pp. 1–12). (In Press). Springer.
- 2 **Sanyal, B., & Sanyal, S. (2026, June).** Traffic Sign Recognition Using Optimized Curvelet Entropy Features. In *the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). (In Press). IEEE.
- 3 **Sanyal, S., Khatua, M., & Chattopadhyay, P. (2026a, June).** CSIDS: Detecting Mimicry Attacks using Network Context with Host Behavior in IoT Networks. In *the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). (In Press). IEEE.
- 4 **Sanyal, S., Priya, T., Debnath, N., Mitra, D., & Mukhopadhyay, S. (2026, June).** Placing Tourist Spots in Smart Cities: A Crowdsourced Game Theoretic Approach. In *the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). (In Press). IEEE.
- 5 **Sanyal, S., Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026b, May).** FIREBOLT: Visitor Frequency Aware Crowdsourced Project Selection for Tourist Spots in Smart Cities using Bucketed Shapley Scores. In *the 14th International Conference on Emerging Internet, Data & Web Technologies (EIDWT-2026)* (Vol. 291, pp. 255–266). Springer. [doi:10.1007/978-3-032-22798-0_23](https://doi.org/10.1007/978-3-032-22798-0_23)
- 6 **Sanyal, S., Khatua, M., & Chattopadhyay, P. (2024, December).** An Adaptive IDS Scheduling Framework for Host Based Mimicry Attack Detection in IoT Networks. In *2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS 2024)* (pp. 1–6). IEEE. [doi:10.1109/ANTS63515.2024.10898203](https://doi.org/10.1109/ANTS63515.2024.10898203)
- 7 **Sanyal, S., Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2024, June).** Crowdsourced Project Selection in Tourist Spots with Voting in Smart Cities. In *Decision Science Alliance International Summer Conference (DSA ISC 2024)* (pp. 105–119). Springer. [doi:10.1007/978-3-031-78241-1_10](https://doi.org/10.1007/978-3-031-78241-1_10)
- 8 **Sanyal, S., Mukhopadhyay, S., Xhafa, F., Singh, V. K., & Sanyal, B. (2022, August).** An Incentive Framework for Surplus Food Distribution in the Context of Smart Cities. In *International Conference on Network-Based Information Systems (NBIS 2022)* (pp. 35–47). Springer. [doi:10.1007/978-3-031-14314-4_4](https://doi.org/10.1007/978-3-031-14314-4_4)

BOOKS AND CHAPTERS

- 1 **Sanyal, S., Mukhopadhyay, S., Xhafa, F., Kumar Singh, V., & Sanyal, B. (2023, May).** Incentives in Surplus Food Distribution for Smart Cities and Beyond: An Activity Aware Solution. In *Advances in engineering and information science toward smart city and beyond* (pp. 75–90). [doi:10.1007/978-3-031-29301-6_4](https://doi.org/10.1007/978-3-031-29301-6_4)

SKILLS

Languages	■ Strong reading, writing, and speaking competencies in English, Hindi, and Bengali.
Coding	■ Python 3, SQL, VBA, C, LaTeX .
Tools	■ Contiki-NG/Cooja Simulator, Visual Studio Code, Arduino IDE, MySQL, MS Excel.

MISCELLANEOUS EXPERIENCE

CERTIFICATIONS AND TRAINING

- 2025 ■ **Active Participation in interactive training session (March 17):** On *Intellectual Property Rights (IPR) Awareness* organized by the *Government of India*, and the *Intellectual Property Rights (IPR) Cell, Indian Institute of Technology (BHU) Varanasi, India*, at the *Indian Institute of Technology (BHU) Varanasi, India*.
- 2024 ■ **Active Participation in three-day International Workshop (July 01 to July 03):** On *Design Automation for Quantum Computing* organized by *IEEE ComSoc Chapter, Guwahati (Kolkata Section)*, at the *Indian Institute of Technology Guwahati, India*.
- **Diploma Degree Sanskrit Linguist (July 2023 to June 2024):** Certified by the *Central Sanskrit University (CSU), Delhi, India*, organized by the *Indian Knowledge Systems (IKS)* in association with the *Indian Institute of Technology Guwahati, India*.
- **Active Participation in thirteen-day Interactive Video Lecture Sessions (April 24 to May 10):** On *Space Science and Technology Awareness Training (START) - Exploration of the Solar System* organized by the *Indian Space Research Organization (ISRO)* and conducted by the *Indian Institute of Technology Guwahati, India*, as a Nodal Centre.
- 2023 ■ **Active Participation in Workshop (November 1):** On *Transitioning a Career in Cyber Security* organized by *Fixity EDX in association with Electronics and ICT Academy of Indian Institute of Technology Guwahati, India*, supported by the *Ministry of Electronics and Information Technology (MeitY), Government of India*.
- **Certification Degree Sanskrit Linguist (July 2022 to June 2023):** Certified by the *Central Sanskrit University (CSU), Delhi, India*, organized by the *Indian Knowledge Systems (IKS)* in association with the *Indian Institute of Technology Guwahati, India*.
- **Active Volunteer and Teaching Assistant in organizing a seven-day high-end Workshop (June 24 to June 30):** On *Power Electronics Hardware Design with Microcontroller Programming* organized by the *Indian Institute of Technology Guwahati, India* sponsored by the *Science and Engineering Research Board (SERB), India* under the *Accelerate Vigyan* scheme.
- 2022 ■ **Active Participation in two-day Workshop (December 19 to December 20):** On *High Performance Computing (HPC)* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *Centre for Development of Advanced Computing (C-DAC), India*.
- **Certified Karate Trainer and Fight Judge (November 20):** *Entry Level Kumite (fight) Judge* certified by the *Karate India Organization (KIO)* affiliated with the *Asian Karate Federation (AKF)* and the *World Karate Federation (WKF)*, recognized by the *International Olympic Committee (IOC)*.
- **Certified Karate Practitioner (November 20):** *Second Dan Black Belt* awarded by the *Traditional Shito-Ryu Karate Do Association (India), TSKA*, approved by the *Karate India Organization (KIO)*, affiliated with the *World Karate Federation (WKF)*, recognized by the *International Olympic Committee (IOC)*.
- **Active Participation in Lecture (November 03):** On *AI-Driven Networks for Earth and Space* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *University of Illinois at Urbana Champaign (UIUC), USA*.
- **Certified Participation in three-day Workshop (August 23 to August 25):** On *Scaling Blockchains: From Bitcoin to the Lightning Network* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *University of Catania, Italy*.
- **Certified Participation in ISEA Virtual Presentation Conclave (IVPC-2022) (June 23):** On *Modeling Attacks on Physically Unclonable Functions (PUFs) and Open Trustless Blockchains: Security Trade-offs* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *Ministry of Electronics and Information Technology (MeitY), Government of India*.

MISCELLANEOUS EXPERIENCE (continued)

- **Active Participation in Webinar (January 14):** On *Healthcare IoT* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *IEEE Communication Society, Technical Committee on eHealth: Special Interest Group on 'IoT for eHealth' and Technical Committee Green Computing and Communications: Special Interest Group on 'Pandemics.'*
- **Certified in Community Services (August 2021 Till Date):** Active participation in social service ventures organized across the state of Assam, India organized by the *National Service Scheme (NSS) cell at the Indian Institute of Technology Guwahati, India.*
- 2021 ■ **Certified Participation in 4.0 Tech Online Bootcamp (September 02):** On *AI and ML, Cyber Security, IoT, Data Science, Augmented Reality (AR) and Virtual Reality (VR), Blockchain, Full Stack* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with *NxtWave, CCBP 4.0 Certification Programs.*
- **Certified Participation in Webinar (August 06):** On *Research Opportunities in Wind Technology* organized by the *Indian Institute of Technology Guwahati, India* in collaboration with *National Institute of Wind Energy (NIWE), Chennai, India.* with the support of the *Ministry of New and Renewable Energy (MNRE), Government of India.*
- 2019 ■ **Certified Karate Practitioner (December 8):** *First Dan Black Belt* awarded by the *Traditional Shito-Ryu Karate Do Association (India), TSKA*, approved by the *Karate India Organization (KIO)*, affiliated with the *World Karate Federation (WKF)*, recognized by the *International Olympic Committee (IOC).*
- 2006 ■ **Certified Swimmer:** Awarded by the *National Swimming Club, Kolkata, India.*

AWARDS AND ACHIEVEMENTS

- 2019 ■ **Participation Award:** Several competitive and charitable running events.
- **Service & Commitment Award:** Recognizing my persistent contributions to the organization.
- 2017 ■ **Merit Award:** Numerous On the Spot awards for outstanding individual contributions.
- 2016 ■ **Service & Commitment Award:** Recognizing my persistent contributions to the organization.
- **Merit Award:** Numerous On the Spot awards for outstanding individual contributions.
- 2015 ■ **Merit Award:** Numerous CLP Faculty awards for distinguished contribution towards knowledge building of associates.
- **Merit Award:** Numerous On the Spot awards for outstanding individual contributions.
- 2014 ■ **Merit Award:** Numerous CLP Faculty awards for distinguished contribution towards knowledge building of associates.
- **Service & Commitment Award:** Recognizing my persistent contributions to the organization.
- 2013 ■ **Merit Award:** Numerous CLP Faculty awards for distinguished contribution towards knowledge building of associates.
- 2011 ■ **Merit Award:** Best Team of the Year in TCS Delhi-NCR region.
- 2006 ■ **Merit Award:** Multiple Maths and Science Olympiads during my schooling.

ACADEMIC INTERESTS

- | | | |
|----------------------|-----------------|---------------------------|
| ■ Internet of Things | ■ Game Theory | ■ Smart devices |
| ■ Machine Learning | ■ Deep Learning | ■ Artificial Intelligence |

NON-ACADEMIC INTERESTS

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|--------------------------|----------------|------------|
| ■ Science and Technology | ■ Martial Arts | ■ Swimming |
| ■ Social activities | ■ Football | ■ Cricket |
| ■ Lawn Tennis | ■ Table Tennis | ■ Running |

ABOUT ME

- Team player with leadership qualities
- Very active and adapt to new settings
- Learn fast, both actively and passively
- Can always get tasks done
- Enthusiastic about interesting work
- Science and technology enthusiast
- Get quality productive very early
- New undone challenges interest me
- Very punctual and adhere to timelines
- Knack to outperform expectations

REFERENCES

Dr. Manas Khatua

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Dept. of Computer Science and Engineering,
Indian Institute of Technology Guwahati,
Guwahati, Assam, India.

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